AN ARTINIAN RING HAS ONLY FINITELY MANY MAXIMAL IDEALS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Chinese remainder theorem for pairwise comaximal ideals. Invoked inside the counted proof at lines 12950-12951. Shows the products of distinct maximal ideals form a strictly decreasing chain of ideals.
- The descending chain condition for ideals in an Artinian ring. Supplies the contradiction at the end of the counted proof (line 12949). Forbids the infinite strictly descending chain of intersections of maximal ideals.

Lemma 0.1. *If R is Artinian then R has only finitely many maximal ideals.*

Proof. Suppose that $\mathfrak{m}_i$, $i = 1, 2, 3, \ldots$ are pairwise distinct maximal ideals. Then $\mathfrak{m}_1 \supset \mathfrak{m}_1 \cap \mathfrak{m}_2 \supset \mathfrak{m}_1 \cap \mathfrak{m}_2 \cap \mathfrak{m}_3 \supset \ldots$ is an infinite descending sequence (because by the Chinese remainder theorem all the maps $R \to \oplus_{i=1}^n R/\mathfrak{m}_i$ are surjective). $\square$

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